Stylianos Loukas Vasileiou

Department of Computer Science
New Mexico State University
Las Cruces, NM, 88003, USA
stelios@nmsu.edu
personal website
Google scholar

RESEARCH INTERESTS

- Human-AI Interaction & Psychology, Collaborative Decision-Making, Computational Argumentation, Cognitive User Modeling, Neurosymbolic AI

EDUCATION

Ph.D. in Computer Science — FL'18 – SP'25
Washington University in St. Louis — St. Louis, USA

M.S. in Artificial Intelligence — FL'16 – FL'17
University of Southampton — Southampton, UK

M.S. in Applied Mathematics — FL'15 – FL'16
University of Glasgow — Glasgow, UK

B.S. in Statistics and Actuarial–Financial Mathematics — FL'10 – FL'14
University of the Aegean — Samos, GR

PROFESSIONAL EXPERIENCE

New Mexico State University — 2025 – now
Assistant Professor, Department of Computer Science — Las Cruces, USA

HONORS AND AWARDS

- *Early Career Award*, International Conference on Principles of Knowledge Representation & Reasoning (KR) — 2026
- *Outstanding Dissertation Award*, International Conference on Automated Planning & Scheduling (ICAPS) — 2026
- *Turner Best Dissertation Award*, Washington University in St. Louis — 2026

FUNDING

Total: $2,324,375.31; as PI: $700,000.

- NASA EPSCoR RID. *Building New Mexico's AI Workforce for Space Weather Forecasting and Nowcasting.* PI: Juie Shetye; Co-PIs: Huiping Cao, **Stylianos Loukas Vasileiou**, and Lipi Mukherjee. $324,376. 2026 – 2028.
- Defense Advanced Research Projects Agency. *Splitting-based Polynomially-bounded Automated Reasoning with Tensorized Adaptation (SPARTA).* PI: Chitta Barral; Co-PIs: **Stylianos Loukas Vasileiou**, Son Tran, Yuliya Lierler, and Jorge Fandino. $1,999,999.31. 2026 – 2028.

PUBLICATIONS

Journal Articles

[1] **Stylianos Loukas Vasileiou**, William Yeoh, Alessandro Previti, and Son Cao Tran. "On Generating Monolithic and Model Reconciling Explanations in Probabilistic Scenarios". In: *Journal of Artificial Intelligence Research (JAIR)* 84 (2025).

[2] **Stylianos Loukas Vasileiou**, William Yeoh, Son Cao Tran, Ashwin Kumar, Michael Cashmore, and Daniele Magazzeni. "A logic-based explanation generation framework for classical and hybrid planning problems". In: *Journal of Artificial Intelligence Research (JAIR)* 73 (2022).

Conference Papers

[1] **Stylianos Loukas Vasileiou** and Olga Derendiaeva. "Discourse-Aware Policy Analysis with Argumentation: A Hybrid LLM-Symbolic Framework for Disaster Governance". In: *Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2026.

[2] **Stylianos Loukas Vasileiou**, Minh Nguyen, Tran Cao Son, Huiping Cao, and Enrico Pontelli. "Extracting Verified Action Theories from Informal Specifications via Explanation-Guided Refinement". In: *International Conference on Principles of Knowledge Representation and Reasoning*. 2026, pp. 1153–1158.

[3] **Stylianos Loukas Vasileiou**, Antonio Rago, Maria Vanina Martinez, and William Yeoh. "Do Humans and LLMs Diverge in Belief Revision? Evidence from a Bayesian Analysis". In: *Conference on Language Modeling (COLM)*. 2026.

[4] **Stylianos Loukas Vasileiou**, Antonio Rago, Francesca Toni, and William Yeoh. "Argumentative Human-AI Decision-Making: Toward AI Agents That Reason With Us, Not For Us". In: *International Conference on Autonomous Agents and Multiagent Systems (AAMAS) (BlueSky Track)*. 2026.

[5] Silvia Tulli, **Stylianos Loukas Vasileiou**, Mohamed Chetouani, and Sarath Sreedharan. "Inferring Implicit Goals Across Differing Task Models". In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2026.

[6] Antonio Rago, **Stylianos Loukas Vasileiou**, Francesca Toni, William Yeoh, and Son Cao Tran. "A Methodology for Incompleteness-Tolerant and Modular Gradual Semantics for Argumentative Statement Graphs". In: *International Conference on Principles of Knowledge Representation and Reasoning (KR)*. 2025.

[7] **Stylianos Loukas Vasileiou** and William Yeoh. "TRACE-cs: A Hybrid Logic–LLM System for Explainable Course Scheduling". In: *International Conference on Principles of Knowledge Representation and Reasoning (KR)*. 2025.

[8] **Stylianos Loukas Vasileiou** and William Yeoh. "TRACE-cs: A Synergistic Approach to Explainable Course Scheduling Using LLMs and Logic". In: *AAAI Conference on Artificial Intelligence (AAAI) (Demo Track)*. 2025.

[9] Yinxu Tang, **Stylianos Loukas Vasileiou**, Vincent Derkinderen, and William Yeoh. "Model Reconciliation via Cost-Optimal Explanations in Probabilistic Logic Programming". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2025.

[10] Yinxu Tang, **Stylianos Loukas Vasileiou**, and William Yeoh. "Does Your AI Agent Get You? A Personalizable Framework for Approximating Human Models from Argumentation-based Dialogue Traces". In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2025.

[11] **Stylianos Loukas Vasileiou**, Ashwin Kumar, William Yeoh, Son Cao Tran, and Francesca Toni. "Dialectical Reconciliation via Structured Argumentative Dialogues". In: *International Conference on Principles of Knowledge Representation and Reasoning (KR)*. 2024.

[12] **Stylianos Loukas Vasileiou**, Borong Xu, and William Yeoh. "A Logic-based Framework for Explainable Agent Scheduling Problems". In: *European Conference on Artificial Intelligence (ECAI)*. 2023.

[13] **Stylianos Loukas Vasileiou** and William Yeoh. "PLEASE: Generating Personalized Explanations in Human-Aware Planning". In: *European Conference on Artificial Intelligence (ECAI)*. 2023.

[14] Ashwin Kumar, **Stylianos Loukas Vasileiou**, Melanie Bancilhon, Alvitta Ottley, and William Yeoh. "VizXP: A Visualization Framework for Conveying Explanations to Users in Model Reconciliation Problems". In: *International Conference on Automated Planning and Scheduling (ICAPS)*. 2022.

[15] Son Cao Tran, Van Nguyen, **Stylianos Loukas Vasileiou**, and William Yeoh. "Model Reconciliation in Logic Programs". In: *European Conference on Logics in AI (JELIA)*. 2021.

[16] **Stylianos Loukas Vasileiou**, Alessandro Previti, and William Yeoh. "On Exploiting Hitting Sets For Model Reconciliation". In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2021.

[17] Son Cao Tran, Van Nguyen, **Stylianos Loukas Vasileiou**, and William Yeoh. "Conditional Updates of Logic Programs and its Applications (Poster Presentation)". In: *International Conference on Principles of Knowledge Representation and Reasoning (KR)*. 2020.

[18] Van Nguyen, **Stylianos Loukas Vasileiou**, Son Cao Tran, and William Yeoh. "Conditional Updates of Answer Set Programming and Its Application in Explainable Planning (Extended Abstract)". In: *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. 2020.

[19] Van Nguyen, **Stylianos Loukas Vasileiou**, Son Cao Tran, and William Yeoh. "Explainable Planning Using Answer Set Programming". In: *International Conference on Principles of Knowledge Representation and Reasoning (KR)*. 2020.

Workshop & Symposium Papers

[1] **Stylianos Loukas Vasileiou** Hopkins Phillip and Sarath Sreedharan. "Can LLMs Explain Plans Well?" In: *International Conference on Automated Planning and Scheduling (ICAPS) Workshop on Human-Aware and Explainable Planning*. 2026.

[2] **Stylianos Loukas Vasileiou**, Antonio Rago, Maria Vanina Martinez, and William Yeoh. "When Humans Revise Their Beliefs, Explanations Matter: Evidence from User Studies and What It Means for AI Alignment". In: *International Conference on Automated Planning and Scheduling (ICAPS) Workshop on Human-Aware and Explainable Planning*. 2025.

[3] Yinxu Tang, **Stylianos Loukas Vasileiou**, and William Yeoh. "Approximating Human Mental Models During Argumentation-based Dialogues". In: *International Conference on Automated Planning and Scheduling (ICAPS) Workshop on Human-aware and Explainable Planning*. 2024.

[4] Silvia Tulli, **Stylianos Loukas Vasileiou**, and Sarath Sreedharan. "Human-Modeling in Sequential Decision-Making: An Analysis through the Lens of Human-Aware AI". In: *International Conference on Automated Planning and Scheduling (ICAPS) Workshop on Human-aware and Explainable Planning*. 2024.

[5] Andrew Estornell, **Stylianos Loukas Vasileiou**, William Yeoh, and Daniel Borrajo. "Predicting Customer Goals in Financial Institution Services: A Data-Driven LSTM Approach". In: *International Conference on Automated Planning and Scheduling (ICAPS) Workshop on Financial Planning*. 2023.

[6] **Stylianos Loukas Vasileiou**. "Towards a Logical Account for Human-Aware Explanation Generation in Model Reconciliation Problems". In: *International Conference on Autonomous Agents and Multiagent Systems (AAMAS) Doctoral Consortium*. 2023.

[7] **Stylianos Loukas Vasileiou**, Ashwin Kumar, William Yeoh, Son Cao Tran, and Francesca Toni. "DR-HAI: Argumentation-based Dialectical Reconciliation in Human-AI Interactions". In: *International Joint Conference on Artificial Intelligence (IJCAI) Workshop on Explainable AI*. 2023.

[8] **Stylianos Loukas Vasileiou**, Ashwin Kumar, and William Yeoh. "On Generating Personalized Explanations via Knowledge Forgetting". In: *International Joint Conference on Artificial Intelligence (IJCAI) Workshop on Explainable AI*. 2022.

[9] **Stylianos Loukas Vasileiou**, William Yeoh, Son Cao Tran, and Alessandro Previti. "Explanations as Model Reconciliation via Probabilistic Logical Reasoning". In: *International Conference on Principles of Knowledge Representation and Reasoning (KR) Workshop on Explainable Logic*. 2021.

[10] **Stylianos Loukas Vasileiou**, William Yeoh, and Son Cao Tran. "On the Relationship Between KR Approaches for Explainable Planning". In: *International Conference on Automated Planning and Scheduling (ICAPS) Workshop on Explainable AI Planning*. 2020.

[11] **Stylianos Loukas Vasileiou**, Son Cao Tran, and William Yeoh. "A Preliminary Logic-based Approach for Explanation Generation". In: *International Conference on Automated Planning and Scheduling (ICAPS) Workshop on Explainable AI Planning*. 2019.

Technical Reports

[1] Hendrik Baier, Mark T Keane, Sarath Sreedharan, Silvia Tulli, Abhinav Verma, and **Stylianos Loukas Vasileiou**. "Explainable AI for Sequential Decision Making (Dagstuhl Seminar 24372)". In: *Dagstuhl Reports* 14.9 (2025), pp. 67–103.

INVITED TALKS

- New Mexico State University, Department of Psychology — Sep. 2026
- ICAPS Launchpad Workshop — Nov. 2025
- Worcester Polytechnic Institute, Department of Computer Science — Feb. 2025
- New Mexico State University, Department of Computer Science — Dec. 2024

ADVISING

Ph.D. Students

- Shahriar Rahman Dipon SP'26 – now
  New Mexico State University
- Subomi Oyewole (co-advised with Juie Shetye, Astronomy) FL'25 – now
  New Mexico State University

TEACHING

New Mexico State University

- CS-5750: Artificial Intelligence II FL'26
- CS-5405: Artificial Intelligence I SP'26
- CS-5996: Explainable AI Decision-Making FL'25

SELECTED SERVICE

Workshop Organizing Committee

- IJCAI Workshop on Explainable AI (XAI) 2025 – now
- ICAPS Workshop on Human-Aware and Explainable Planning (HAXP) 2021 – now

Senior Program Committee

- International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2027

Program Committee

- Annual Meeting of the Association for Computational Linguistics (ACL) 2026 – now
- International Conference on Principles of Knowledge Representation and Reasoning (KR) 2026 – now
- International Joint Conference on Artificial Intelligence (IJCAI) 2023 – now
- Association for the Advancement of Artificial Intelligence (AAAI) 2024 – now
- International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2023 – now
- International Conference on Automated Planning and Scheduling (ICAPS) 2024 – now
- International Conference on Learning Representations (ICLR) 2025
- Neural Computing and Applications (NCAA) 2025 – now
- European Conference on Artificial Intelligence (ECAI) 2023, 2025

External Reviewer

- Artificial Intelligence Journal (AIJ) 2023 – now